

Drain Detectives

Introduction:

Students will investigate storm drains around their school, document each drain's condition and nearby filtration features, identify the waterway it might connect to, and combine their findings into a class map.

Learning Objectives:

- Explain what a storm drain is and how it differs from a household drain
- Explain how stormwater runoff can carry pollution directly into lakes, rivers, and streams
- Identify four Low Impact Development (LID) features: rain gardens, bioswales, permeable pavement, and native plant strips
- Collect and record observations about storm drains around the school
- Use collected evidence to identify a drain that should be prioritized for improvement

Materials (Teams of 4-5 students):

Material	Quantity
Large printed map of the school and surrounding area	1pc
Green and red pins, stickers, or markers	As needed
Student recording sheet or notebook	1
Pencil or pen	1-2
Phone, tablet, or camera for photographs	1

Instructions:

1. Present slides 1-4 to give students an introduction to what storm drains are and context for the activity.
2. On the "Your Task" slide explain that students will locate drains around the school and determine whether filtration systems as per the slide's instructions
3. Review all seven pieces of information student should collect for every drain
 - a. Drain number
 - b. Location
 - c. Connected waterway
 - d. Condition
 - e. Low Impact Development feature (Yes or No and which)
 - f. Estimated distance to the nearest body of water
 - g. Photograph
4. Provide one model one entry before students go outside and present the various tools to use to locate bodies of water ("Connecting Drains to Streams")

5. Bring students back inside and build a class map using the large printed map of the school. Have teams add their drains one at a time and design the map according to the instructions on the slide ("Building the Class Map").
6. Use the completed map to discuss the following questions
 - a. How many drains did the class find in total?
 - b. How many drains are unfiltered? How many have LID features?
 - c. Which streets or areas have the highest concentration of unfiltered drains?
 - d. What specific improvement or policy should be recommended?

Data Sheet:

Student Name(s):	
Team Number:	
Assigned Area:	

[illegible]